

Nathaniel Billig

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EDUCATION

University of Wyoming , College of Engineering and Physical Sciences	Expected May 2027
Master of Science: Electrical Engineering	Cumulative GPA 3.9/4.0
Relevant Coursework: <i>Advanced Robotics Modeling, Robot Learning, Robot Perception, Cooperative Robotics, AI for Multiagent Systems,</i>	
University of Wyoming , College of Engineering and Physical Sciences	May 2025
Bachelor of Science in Computer Engineering; Minors in Computer Science and Mathematics	Cumulative GPA 3.8/4.0
Relevant Coursework: <i>Advanced Microprocessors, HDL Digital Design, Electronics, GPU Programming for HPC, Computer Security, Neural Networks, Deep Reinforcement Learning</i>	

EXPERIENCE

Graduate Robotics Researcher — USSF SSTI Sponsored Research	Sept 2024 - Present
University of Wyoming	
- Led the mechanical and electrical development of a tendon-driven, underactuated robotic gripper for non-cooperative object grasping, integrating CAD, embedded control, custom PCB electronics, and multi-sensor feedback - Iteratively refined the mechanical and electrical design based on experimental testing, performance results, and observed system limitations - Developed simulation and calibration methods to characterize gripper kinematics, joint sensing, and contact forces	
Microcontroller Project Engineer, 4th State Communications - Cheyenne, WY	Oct 2023 - Feb 2024
- Developed C/C++ Arduino firmware for UART communication and control of telecommunications transmitters - Built Python software using Linux NFS to transmit control commands between a central controller and networked relay systems	
Undergraduate Research Assistant, University of Wyoming - Laramie, WY	May 2023 - Sept 2025
- Developed computer vision applications and embedded Python/C/C++ software for Arduino and Raspberry Pi-based autonomous platforms - Instructed high school students and teachers in physical computing, programming fundamentals, and hands-on electronics	
Supplemental Instruction Leader, University of Wyoming - Laramie, WY	Dec 2023 - May 2024
Organized study sessions and created instructional problem-solving worksheets for ordinary differential equations	

PROJECTS

Hall Effect Robotic Joint Angle Sensing , University of Wyoming
· Integrated Hall-effect sensors and magnets into robotic joints, developing embedded software and calibration methods for real-time joint-angle measurement and feedback
Multi-Agent Continuum Arm Q-Learning , University of Wyoming
· Developed a multi-agent Q-learning framework for coordinated control of a continuum robotic arm, evaluating discrete state and action spaces for joint coordination and motion control
Contact Force Simulation , University of Wyoming
· Developed a MATLAB/Simscape model to reproduce robotic gripper kinematics and estimate contact forces at the finger-object interface, comparing simulated motion with experimental results
Digital Twin Recreation , University of Wyoming
· Recreated a physical robotic gripper as a digital model to represent its mechanical structure and motion, enabling system visualization and comparison between simulated and physical behavior

SKILLS

Programming & Embedded Systems: C/C++, Embedded Programming, Python, MATLAB, Arduino, Raspberry Pi
Electronics & Communication: UART, SPI, I ² C, PCB/Schematic Design, Sensor Integration,
Robotics & Simulation: ROS, Kinematics, Path Planning, Digital Twins, Reinforcement Learning, Computer Vision
Software & CAD: SolidWorks, Fusion 360, OpenCV, MATLAB/Simulink, Simscape, Microsoft Office Suite